Food Fight: U.S. Exporters' Adjustments to Russia's 2014 Agricultural Import Ban

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Abstract

This paper examines the impact of Russia's 2014 food-import ban on U.S. firms that exported banned products to Russia. Using confidential customs transaction data, we implement triple-difference and dosage-response approaches to identify how firms adjust to the sudden loss of a market. Following the ban, treated firms experienced a 30 percentage-point decrease in the probability of exporting banned food to Russia relative to control firms. However, there is substantial heterogeneity by pre-ban reliance on the Russian market: heavily reliant firms were significantly less likely to survive once the ban was in place, and survivors experienced large reductions in revenue (19%) and total export value (49%) for each standard deviation increase in Russian market exposure. We find evidence of export redirection to neighboring countries, though it is insufficient to offset losses. Any negative impacts on survivors dissipate by five years post-ban.

Keyword: sanctions, import bans, U.S.-Russia trade

JEL Classification: F51, F13, F14, L66

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1 Introduction

Restrictions on trade, which range from regulatory policies to outright bans, impose varying degrees of economic costs on trading firms. Growing geopolitical tensions have increased trade sanctions worldwide and are likely to continue to intensify their use (Yotov, Yalcin, Kirilakha, Syropoulos, and Felbermayr, 2020; Gopinath, Gourinchas, Presbitero, and Topalova, 2025). Thus, understanding firm responses to sanctions-induced trade restrictions is important to minimize impacts on domestic economic activity. This paper studies the impact of a 2014 Russian import ban on select food and agricultural products on U.S. food exporters.

The U.S. agricultural sector relies heavily on exports. About a fifth of U.S. agricultural output is exported, making the U.S. the second-largest agricultural trader in the world, behind the European Union (U.S. Department of Agriculture, 2018, 2025). Consequently, the sector is very sensitive to trade barriers. The U.S. agricultural sector maintained a trade surplus for almost sixty years until 2019, but has experienced a growing trade deficit since then due in part to rising trade barriers (Kaufman, 2025).

To understand the impact of trade restrictions on the highly export-dependent agricultural sector, we study an extreme trade barrier: an outright ban imposed by Russia in 2014 on the import of select U.S. agricultural products. This ban was part of a tit-for-tat between the United States and its Western allies, on the one hand, and Russia, on the other, following Russia's invasion of the Ukrainian region of Crimea in February 2014. The U.S. and its allies responded to the invasion with a series of sanctions covering individuals, capital markets, and exports of specific goods and services. On August 6, 2014, Russia retaliated by imposing a ban on some food imports from the United States, European Union, Canada, Australia, and Norway (U.S. Department of Agriculture, 2014a).

In 2013, prior to the imposition of the ban, Russia was the 20th largest export destination for U.S. agricultural and related products, although Russia's importance as an export market also varied by product: it was the fourth largest market for U.S. poultry and eighth largest for U.S. fish and seafood, but only the 19th largest market for U.S. fresh fruit (U.S. Department

of Agriculture, 2014b). The import ban affected approximately 55% of the total value of U.S. exports of agricultural goods to Russia in 2013.[1]

Russia's action had an immediate and dramatic impact on U.S. food exports to Russia: exports of banned food from the U.S. to Russia declined by approximately 90% right after the imposition of the ban, and continued to decline in the following quarter. Figure 1(a) shows this sharp decline in food exports to Russia relative to exports of the same products to two control countries: China and Angola.[2] A decomposition of the decline in the aggregate value of banned-food exports into changes in quantity and changes in unit values, reported in Appendix A.3, shows that the impact operates entirely through a change in quantity of exports. Following this decline, the value of U.S. banned-food exports to Russia remained depressed at least through 2019. This pattern is consistent with the marked increase in exports of banned food to Russia from non-sanctioned exporters in the aftermath of the ban (Bělín and Hanousek, 2021). In contrast, exports to the control countries exhibit a much smaller decline in early 2015, and export levels remain much higher through 2019. Non-banned food exports to Russia exhibited a smaller, and much more gradual, decline, as shown in Figure 1(b).[3]

To understand how the Russian food-import ban impacted U.S. exporters, we use confidential customs transaction-level data from the U.S. Census Bureau. We identify treated firms as U.S. exporters that had previously sold banned food to Russia and examine their margins of adjustment to the near complete loss of a market. Specifically, we ask: how does the loss of a market impact firm survival and overall firm performance? To what extent do exporters that have lost a market increase their exports to other destinations, increase their exports of other products, or do neither and contract? We answer these questions using

[1] Authors' calculations using U.S. Census Bureau (2022). Appendix Table A-1 reports the complete set of banned-food products, which include meat, fish, dairy, vegetables, fruits and nuts.

[2] We select these control destinations using a synthetic-control approach described in Appendix A.2.

[3] Food products not subject to the ban are defined as any products in Schedule B Chapters 1–24, excluding the specific banned products. Non-banned food products include live trees and plants (Schedule B Chapter 6), coffee, tea, and spices (Chapter 9), and tobacco products (Chapter 24).

a triple-difference (DDD) specification. We compare the exporting behavior of these U.S. firms in the pre-ban period to their exporting behavior once the ban was in place (the first difference). Our first set of control firms also exported to Russia in the pre-ban period but did not export any banned products — neither banned food nor any of products subject to U.S. export restrictions — to Russia (the second difference). Finally, because food exports are subject to various supply-side shocks due to weather and product yields, we include a further comparison with firms exporting food and other products to China and Angola (the third difference).

We supplement the DDD empirical design with "dosage-response" regressions to account for the significant heterogeneity in exporting firms' exposure to the Russian market. These regressions use only firms that exported banned-food product to Russia prior to the implementation of the ban, exploiting variation in firms' exposure to the Russian market in 2012–2013. In addition to allowing us to examine the differential impact of the ban by a firm's relative reliance on the Russian market, these specifications enable us to study outcomes that cannot be examined using the DDD regressions; specifically, we can separately examine whether firms diverted exports to Russia's close neighbors or to more distant destinations.

Consistent with the decline in aggregate exports of banned food to Russia following the imposition of the ban, the food-import ban had an abrupt and permanent effect at the firm level. Using the DDD framework, we find a sharp decline of 30 percentage points (pp) in treated firms' likelihood of exporting banned food to Russia relative to control firms. Because control firms also exhibit a decrease, this relative 30pp decrease does not represent the full absolute decrease in trading probability for treated firms. Instead, the DDD estimates capture the larger decline in the treated firms' extensive margin of exporting banned products to Russia, relative to the control firms' decline in the probability of exporting their respective products to their respective destinations.

Treated firms partially compensated for the loss of the food market by exporting other,

non-banned, products to Russia, and, on average, appear to have suffered no adverse effects from Russia's import ban. However, these results mask substantial heterogeneity in firms' responses. Firms that were relatively more reliant on the Russian export market in the pre-ban period were more likely to cease operations entirely: a one standard-deviation increase in reliance on the Russian market is associated with a 10pp decrease in a firm's probability of remaining in operation. Even conditional on survival, a one-standard-deviation increase in Russian-market exposure is associated with a 19% larger reduction in revenue and a 49% larger reduction in total export value. In contrast, we find no evidence of meaningful differential reductions in employment or payroll. Five years after the imposition of the ban, any remaining negative effects among survivors become statistically insignificant, suggesting that firms that survived were able to find new markets or other revenue streams.

The key contribution of this paper is a quantification of U.S. firms' responses to the sudden and complete stop of sales in a heavily export-oriented sector to an important foreign market. This paper relates to prior work examining impacts of the 2014 Russian sanctions, specifically the food-import ban, on firms in other Western countries. Crozet and Hinz (2020) find that the impact of the ban on French firms extended beyond the specifically targeted products to a general chilling effect on trade with Russia. Gullstrand (2020) finds that the ban caused a large decline not only in Swedish firms' exports to Russia, but also in sales to other foreign markets that similarly faced Russian food-import ban. Crozet, Hinz, Stammann, and Wanner (2021) find a sharp decline in the probability that a French firm trades with Russia after the imposition of both the European and Russian sanctions. Lastauskas, Proškutė, and Žaldokas (2023) study the impact of the Russian food-import ban on Lithuanian firms that exported food to Russia in 2013. Similar to findings in this paper, they find that exporters with a high dependence on the Russian market were more likely to find alternative export markets in the aftermath of the ban. Finally, Kohl, van den Berg, and Franssen (2023) find that Dutch firms experienced a decline in both the intensive and

extensive margin of exports in response to the Russian counter-sanctions.[4]

Our paper provides the first examination of the Russian food-import ban's impact on U.S. exporting firms. This focus offers insights into U.S. firm adjustments in a sector that has historically been susceptible to foreign actions due to its reliance on foreign export markets. For example, in 2018 amd 2019, the U.S. government increased tariffs on imports from various countries. Several trade partners imposed retaliatory tariffs on many U.S. agricultural and food products, such that average foreign tariffs on these products increased from eight to almost 29 percent (Carter and Steinbach, 2020). In March 2025, Canada and China imposed import tariffs on a wide range of U.S. agricultural goods including wheat, corn, cotton, sorghum, soybeans, poultry, pork, and beef, in response to U.S. tariffs imposed earlier in the year (Rosch and Tsui, 2025).

The rest of the paper is organized as follows: Section 2 describes the confidential micro-data sources. Section 3 lays out our twin identification strategies. Results are presented in Section 4. Section 5 concludes.

2 Data

We use the Longitudinal Firm Trade Transactions Database (LFTTD), which includes transaction-level detail on the universe of U.S. merchandise export shipments valued over US$2,500 beginning in 1992 (Kamal and Ouyang, 2020).[5] The LFTTD contains a firm identifier associated with each export transaction, enabling us to link shipments to different destinations to a single firm, and also to link the LFTTD to the Longitudinal Business

[4]Less directly related to our work are papers studying the impact of the conflict on Russian firms. These papers focus on the impact of U.S. and European Union sanctions that targeted specific Russian individuals and firms. Some examples include Ahn and Ludema (2020), who find a negative effect of these sanctions on Russian firms' revenues, and Nigmatulina (2022), who finds that Russian firms' revenue *increased* following sanctions, but that this effect is entirely attributable to increased subsidies to state-owned sanctioned firms.

[5]We use the e201901 version of the LFTTD.

Database (LBD), discussed below. For each exported shipment, we observe the month and year of export, the destination country, and the product traded (ten-digit Schedule B code).

We identify banned products by their Schedule B codes, using a translation provided by the U.S. Department of Agriculture of the original Russian decree listing Schedule B product ranges banned by Russia (U.S. Department of Agriculture, 2014a). These cover poultry, pork, beef, tree nuts, fresh and processed fruits, prepared foods, and fresh vegetables; Appendix Table A-1 lists the Schedule B codes targeted by the Russian sanctions. The banned food products accounted for a little more than half of the value of food and agricultural exports to Russia in 2013 (authors' calculations from U.S. Census Bureau, 2022). Next, we identify U.S. exporters affected by the Russian food-import ban as firms that exported banned food to Russia in 2012 and/or 2013. There are approximately 300 U.S. firms affected by the ban. These are the treated firms (T).[6]

Our analysis is at the firm-year-product-destination level. Because the ban was imposed in early August, and because we are concerned about anticipatory effects during Russia's amassing of troops on the border with Crimea and during the invasion and annexation, but before formal sanctions were imposed, we make the following adjustment. We include 2012 and 2013 as calendar years, but we drop export shipments between January and July 2014. Each post-ban year runs from August through July of the following calendar year. Thus, the analysis sample consists of seven years of data for each firm-product-destination triple: 2012, 2013, and post-ban years 2014/15, 2015/16, 2016/17, 2017/18, and 2018/19. We begin the analysis in 2012.

We add annual information from the Longitudinal Business Database (LBD) on each firm's employment in the pay period that includes March 12 and its calendar-year payroll

[6]All counts are rounded to comply with Census Bureau disclosure-avoidance rules. We drop from our analysis any firm that exported both food and products subject to U.S. export controls to Russia in 2012–2013 to ensure that we capture the impact of the Russian import ban only.

and revenue.[7] We use firms' total value of exports in the LFTTD as another measure of size. Because payroll and revenue are annual, we drop 2014 entirely from the firm-size analysis to avoid confounding the "post-ban" period with any anticipatory effects.

Table 1 shows summary statistics for the treated firms in 2012 and 2013. By construction, every treated exporter had at least one shipment of banned food to Russia in 2012 or 2013; but in any one of those years, only 75% of treated exporters had at least one such shipment. In addition to banned food, on average across 2012 and 2013, 18% of treated firms also exported non-banned products to Russia.

The treated firms are fairly diversified in their destination markets. On average in 2012 and 2013, 86% had at least one banned-food shipment to a country other than Russia, and virtually all of these firms exported these products to at least one destination outside the neighboring set of countries of Belarus, Poland, Latvia, and Lithuania; only 22% also shipped banned food to one of those neighboring countries.

Banned food forms a small but non-negligible part of treated firms' portfolios in the pre-ban period. On average across treated exporters, exports of food to Russia accounted for 15% of total export value, with a standard deviation of 28%.

Finally, treated firms were large, employing on average about 2,300 workers in the pre-ban period, and their average payroll well exceeded $100 million (in 2019 dollars). Average export value was $130 million, and total revenue averaged over $2 billion.

[7] The construction of the LBD is described in Chow, Fort, Goetz, Goldschlag, Lawrence, Perlman, Stinson, and White (2021). We use version v201900 of the LBD for all outcomes except for 2019 revenue, which we obtain from v202100. Because only approximately 80% of firm-year observations have revenue information (Haltiwanger, Jarmin, Kulick, and Miranda, 2015), we use propensity-score weighting to adjust for nonrandom missing revenue. For more information about revenue in the LBD, see Haltiwanger, Jarmin, Kulick, Miranda, Penciakova, and Tello-Trillo (2019).

3 Empirical Approach

Difference-in-Difference-in-Difference Regressions

We implement a triple-difference (DDD) identification strategy. The intuition for the identification is as follows. Treated (T) firms are those that exported banned food to Russia in 2012 and/or 2013. We compare treated firms' outcomes to firms that also exported to Russia in 2012 and/or 2013, but whose export portfolios to Russia excluded banned food. This set of control firms (C1) is subject to the same exchange-rate and demand shocks that are specific to the Russian economy, as well as any overall reduction in Russian demand for U.S. goods due to geopolitical tensions. Firms in C1 include exporters of non-banned food products, as well as exporters of non-food products to the Russian market. We exclude firm that exported products subject to U.S. export controls from both T and C1.

A difference-in-difference (DD) specification with T and C1 firms would remove any common shocks and identify how the change in sanctioned firms' outcomes between the pre-ban period (2012–2013) and the post-ban period (mid-2014 to mid-2019) differs from the change in the non-banned firms' outcomes. However, firms that previously exported banned food may be subject to very different types of demand and cost shocks from firms that exported only non-banned products. We add two additional sets of control firms to address this concern: those that exported banned food to selected other countries (C2), and those that exported only non-banned products to those same selected other countries (C3). Intuitively, the DDD regression is a DD regression that compares T and C1 firms, and then differences out the difference between C2 and C3 firms. The firms in C2 experience the same product-specific cost and demand shocks as the treated firms, and the firms in C2 and C3 share common exchange-rate and country-specific shocks.

To select the control countries, we use public-use data at the monthly level on export value by product and destination. Our method is described in Appendix A.2. The control countries are China and Angola.

We have approximately 300 treated firms, 6,300 firms in control group C1 (exported only non-banned products to Russia in 2012–2013), 1,600 firms that exported banned food to China and/or Angola in 2012–2013, but did not export any products to Russia in those years (C2), and 40,000 firms that exported only non-sanctioned products to China and/or Angola in the same period, and likewise did not export any products to Russia (C3). The restrictions on control firms ensure that they are not also members of the treated set.

We estimate the following specification:

$$\mathrm{Y}_{it} = \alpha_i + \gamma_{jt} + \delta_{dt} + \eta_{jd} + \beta \cdot \mathrm{Post}_t \cdot \mathrm{Treated}_i + \varepsilon_{it} \tag{1}$$

where i designates a firm and t a year. Each firm i is associated with a treatment status $j(i)$ (exporter of banned products or only non-banned products in 2012 and/or 2013) and a destination $d(i)$ (exported to Russia or only to a control country in 2012 and/or 2013). The firm fixed effects α_i capture firm-specific, time-invariant differences in the outcome variable Y_{it}. There are two sets of year fixed effects. The first, γ_{jt}, is a year-specific fixed effect for each treatment status; i.e., one set of year fixed effects common to the treated firms and firms that exported the banned product to the control countries in 2012 and/or 2013 (T and C2), and another common to firms that exported only non-banned products in 2012 and/or 2013 (C1 and C3). These are intended to capture industry-level dynamics shared by exporters of similar products. Similarly, δ_{dt} is a year fixed effect for each destination (T and C1; C2 and C3), intended to capture macroeconomic and policy differences across the two destinations, for example due to Russia's accession to the World Trade Organization in August 2012, exchange rate fluctuations, etc.[8] Finally, η_{jd} are product-destination fixed effects; these are perfectly collinear with the firm fixed effects α_i.

The coefficient β captures differential change in the outcome variable for treated firms

[8]By differencing out any reduction in demand that is common across all Russian imports of U.S. goods, we effectively remove the chilling effect estimated by Crozet and Hinz (2020).

in the post-ban period. The main identifying assumption is that of "parallel trends": but for the imposition of the Russian import ban, the differential trend in the exports of firms selling food to Russia and those selling non-banned products to Russia would have been similar to the differential trend for firms exporting to China and Angola.

We explore the extensive margin of firm activity using an indicator variable for the firm having any payroll, revenue, or exports.[9] We consider a firm to be active if it has positive payroll, revenue, or exports. We also report results from examining the intensive margin of firm employment, payroll, revenue, and exports; these regressions use logged values as outcomes and therefore restrict the sample to observations with nonzero outcomes.

We also explore the extensive margin of exports for a firm-product-destination-year tuple. We generally refer to the extensive margins of exporting as, e.g., "indicator for any shipments of banned food to Russia." This describes the way the indicator is defined for the treated firms. The variable is defined slightly differently for each control group, depending on the focal product and focal destination for that sample. For example, for sample C1, it is an indicator for any shipments of *non-banned products* to Russia; for sample C2, it is an indicator for any shipments of banned food to China and/or Angola; and for sample C3, it is an indicator for any shipments of any non-banned products to China and/or Angola, because the sample is defined. Similarly, "indicator for any shipments of banned food to destinations other than Russia" is "indicator for any shipments of non-banned products to destinations other than Russia" for sample C1.

To test whether there are differential pre-existing trends for treated and control firms and to analyze the dynamic impact of the ban, we also estimate event-study specifications. We focus these specifications on two outcomes: any exports to Russia, and any exports of

[9]Because revenue, payroll, and exports are each obtained from a different data source, it is possible for a firm to have only one or two of these in our analysis dataset.

banned food to Russia. The regression is:

$$\mathrm{Y}_{it} = \alpha_i + \gamma_{jt} + \delta_{dt} + \eta_{jd} + \beta_t \cdot \mathrm{Year}_t \cdot \mathrm{Treated}_i + \varepsilon_{it} \tag{2}$$

where Year_t is a year fixed effect, and Year_{2013} is the omitted time period. The coefficients β_t trace the dynamic effect of the ban on the outcome variable.

Dosage-Response Regressions

The DDD design implicitly assumes that all treated firms respond similarly to the ban. We adopt an alternate identification strategy — dosage response — to assess how firms more reliant on banned-food exports to Russia responded to the import ban compared to firms less reliant on banned-food exports to Russia. Using only the treated firms, we estimate a difference-in-dosage response:

$$\mathrm{Y}_{it} = \alpha_i + \delta_t + \lambda \cdot \mathrm{Post}_t \cdot \mathrm{Dosage}_i + \varepsilon_{it} \tag{3}$$

where Dosage_i is the share of a firm's 2012–2013 exports that are accounted for by sales of banned product.

This specification does not rely on any of the assumptions in the DDD specifications. Instead, within the set of treated firms, we use "low-dosage" firms as controls for "high-dosage" firms. Consequently, the interpretation of the coefficient λ is not directly comparable to the β coefficients from the DDD and event-study specifications. To interpret these estimates, we scale λ by the standard deviation of dosage from Table 1 to calculate how much larger the effect of the ban is for treated firms with relatively more versus less exposure to the Russian food market.

Because this specification relies only on treated firms, it also allows us to estimate the impact of the ban on outcomes that are defined only for these firms. Specifically, we examine the probability that treated firms export banned food to neighboring countries

of Belarus, Poland, Latvia, and/or Lithuania. We refer to these countries collectively as "neighboring countries," even though the actual set of neighboring countries is much larger.[10] A comparable analysis is infeasible in the DDD specification because the synthetic-control countries do not have equivalent neighbors.

We also estimate event-study regressions for the dosage sample:

$$\mathrm{Y}_{it} = \alpha_i + \delta_t + \lambda_t \cdot \mathrm{Year}_t \cdot \mathrm{Dosage}_i + \varepsilon_{it} \tag{4}$$

where, as above, Year_{2013} is the omitted time period. The coefficients λ_t allow us to trace the speed and duration of the impacts. We use these event-study specifications to examine the dynamic effects of the ban on overall firm performance: employment, payroll, revenue, and total value of exports.

4 Results

Firm Survival and Export Patterns

We begin by estimating the impact of the Russian food-import ban on the survival of the treated firms before turning to their likelihood of exporting in Table 2. Panel A shows DDD estimates, and Panel B shows estimates from dosage-response models. The first column in Panel A displays the impact of the ban on "any activity" which is an indicator that equals 1 if the firm has any payroll, revenue, or exports.[11] The positive, but small (4%) and marginally significant DDD coefficient indicates that the ban did not adversely impact treated firms' survival compared to control firms. However, conditional on survival, column 2 shows that the ban reduced the probability that treated firms continued to export banned food to Russia

[10] The full set of geographic neighbors also includes Azerbaijan, China, Estonia, Finland, Georgia, Japan (maritime border), Kazakhstan, Mongolia, North Korea, Norway, Ukraine, and the United States (maritime border).

[11] Over the entire sample period, the mean of the activity indicator was 88.3% for treated firms.

by 30pp relative to the control group. Recall from Table 1 that the mean probability that treated firms' export of banned food to Russia prior to the ban taking effect was 75%, so the 30 percentage-point (pp) decline represents a 40% decline relative to this baseline.

To square this result with Figure 1(a), which shows a 90% decrease in the value of banned-food exports to Russia using public data, note that the treated firms' probability of exports of banned food to Russia falls, mechanically, starting in 2014. This is because the firms are selected into our sample with the condition of having some exports of targeted products to the focal country in the pre-treatment period. Over time, firms may change their export-product mix and destination countries, or they may go out of business altogether. The same is true for each of the three control groups C1, C2, and C3. This coefficient estimate captures the larger decline in the probability of exporting banned food by the treated firms to Russia compared to the decline in the probability of exporting banned food and other products by the control firms.

Figure 3 confirms that there are no pre-ban trends and that the impact is sharp and permanent. Panel (a) shows event-study coefficients from a regression where the outcome variable is an indicator for the firm having any exports to Russia, and panel (b) shows the coefficients when the outcome variable is an indicator for any banned-food exports to Russia. Not surprisingly, the impact of the ban appears sharp and permanent for both outcomes and larger for exports of banned food to Russia.

In contrast to the null effect on firm survival in the DDD specification, the dosage-response estimates, in column 1 of Panel B, show a large and negative effect of the ban on firm survival. To interpret this coefficient, recall that the standard deviation of the dosage measure is 0.28 (from Table 1). Thus, a one-standard-deviation increase in exposure to the ban reduces a treated firm's probability of having any economic activity by 0.3821×0.28, or more than 10pp. This result underscores the heterogeneity in the impact of the ban that varies with a firm's reliance on exporting banned food to the Russian market prior to the ban.

Columns 3-5 in Panel A show the probability of export of other products and/or to other destinations. On average, the indirect effect of the ban on firms' probability of exporting non-banned products to Russia (column 3) is an increase of about seven percentage points. In contrast, the estimated effect on exports to other destinations (columns 4-5) are economically small and statistically insignificant.

The dosage-response regressions in Panel B enable a focus on specific destinations. Column 2 of Panel B shows our estimate of the differential impact of the ban on firms with a higher reliance on the Russian export market. There is no *ex ante* reason to expect that the extensive margin — whether a firm exports banned food to Russia — would vary with the firm's prior dependence on the Russian market. Nevertheless, we do find a negative and marginally significant coefficient, implying that a one standard-deviation increase in exposure to the ban reduces a treated firm's probability of exporting banned food to Russia in the post-ban years by approximately 2.5pp. We interpret this relatively small estimate to imply that the ban was so effective and universal that it affected all exporters to Russia more-or-less uniformly, irrespective of the degree of reliance on the Russian market.[12]

The next two columns of Panel B convey an interesting story: increased reliance on the Russian banned-food market is associated with a modest increase in the probability of exporting banned food to Belarus, Poland, Latvia, and Lithuania (column 3), but not with an increase in exports to other destinations (column 4). The first coefficient suggests that firms that were more reliant on the Russian market may have more successfully leveraged their presence in the region to introduce or expand sales of other products as a way of mitigating the effect of the ban. Smuggling banned food into these neighboring countries for export has been noted as a potential circumvention strategy (Liefert and Liefert, 2015). Our finding is consistent with this concern, but there are also other reasons why exporting

[12]The negative sign may reflect the fact that in the pre-ban period, firms with a high dependence on the Russian market were likely to export banned food to Russia in both 2012 and 2013, whereas firms with a lower dependence on Russia may have banned food to Russia in only one of the two pre-ban years.

banned food to Belarus, Poland, Latvia, and Lithuania may be easier than exporting to other destinations for firms that already have experience exporting to Russia: they share a common language, and both institutions and consumer preferences may be similar.

Firm Activity

We examine select intensive-margin outcomes in Table 3. All outcome variables are logged, and condition on the firm having some activity. The DDD results in Panel A indicate no economically meaningful impact of the ban on average employment, payroll, or revenue of treated firms.[13] However, we find a positive, large (10%), but statistically insignificant effect of the ban on export value.

To determine whether this average effect masks heterogeneous responses by firms more or less reliant on Russia, Panel B shows dosage-response estimates for the same four firm-performance measures. In this panel, all coefficients are negative and two of them — in the revenue and export regressions — are statistically different from zero. Again, referring to the standard deviation of the dosage variable from Table 1, a one-standard deviation increase in a firm's reliance on banned-food exports to the Russian market is associated with a 19% decrease in annual revenue and a 49% decrease in export value on average over the first five post-ban years, even conditional on the firm continuing to have positive economic activity.

We also estimate dosage-response event-study regressions for the four continuous firm-performance outcomes. Figure 2 shows the sign and significance of the event-study coefficients λ_t. The smaller and lighter the marker, the less statistically significant it is; the larger and darker markers signify coefficients that are statistically different from zero. Panel (a) shows the employment coefficients; consistent with the very small and statistically insignificant effect estimated in Table 3, the post-ban coefficients are all statistically insignificant and there is no discernible pattern in their signs. Panel (b) shows the payroll coefficients. All

[13]The logged regressions are estimated using approximately 88% of the sample based on the treated firms' average survival probability. However, the exact samples vary by outcome variable.

but one of these is statistically insignificant, but all the post-ban coefficients are negative, consistent with a mild negative impact on firm payroll. In contrast, panels (c) and (d) show the revenue and export-value coefficients, several of which are statistically significant at the 1% level. In the case of revenue, the negative coefficient is most significant in the year immediately after the ban was imposed and moderates thereafter. In the case of export value, the coefficient is most significant in the second and third years after the ban was imposed and moderates thereafter.

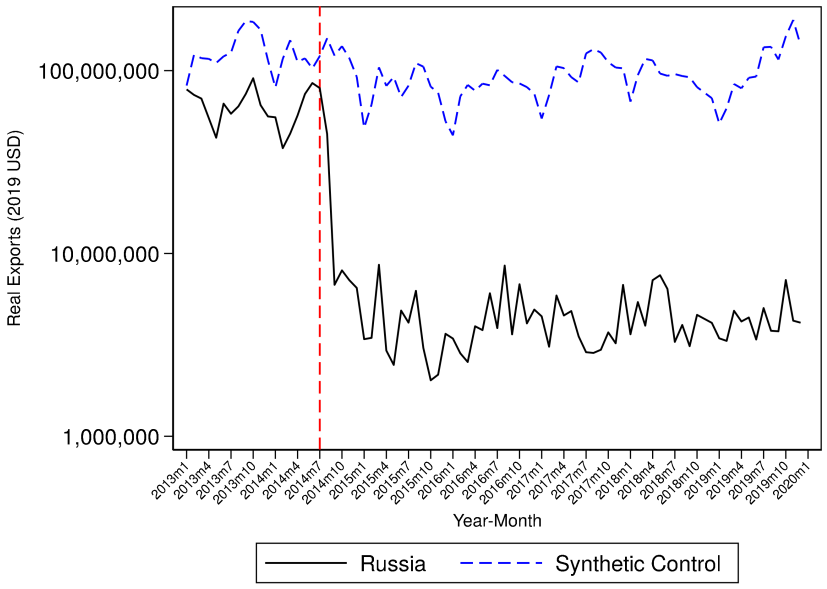
Combined, the DDD and dosage-response results paint a nuanced picture of the impacts of the Russian import ban. For less-reliant firms — those that exported some banned food to Russia, but for which these exports were only a small part of the export portfolio — the impact of the ban was limited and maybe even positive. These firms may have been spurred to make investments in other product and export markets. Their probability of survival consequently increased, as did the value of their exports conditional on survival. In contrast, for firms with a high reliance on banned-food exports to Russia, the impact of the ban was negative and very large. These firms were less likely to survive after the ban was put in place and, even conditional on survival, experienced large decreases in revenue and export value.

5 Conclusion

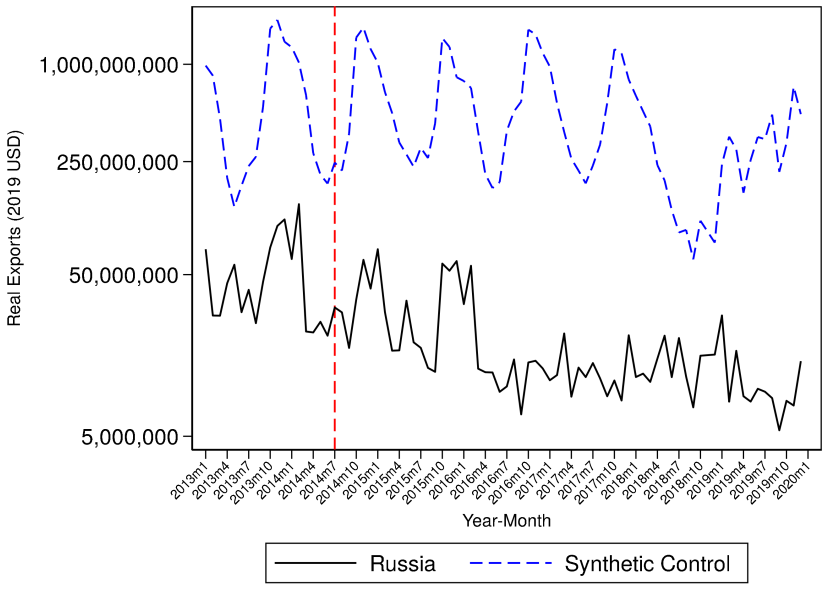
U.S. firms that export to target markets are the main casualties when sanctions impede sales to these markets. This paper examines how U.S. exporters responded to the complete loss of the Russian market for some food and agricultural products in 2014 following Russia's food-import ban. In complementary analyses, we use DDD and dosage-response regressions to estimate both the average response to the sanctions and the presence of heterogeneous responses. We find that the average effect of the sanctions was small, but that firms that relied heavily on the Russian market for banned food experienced significant negative effects for several years. These firms' probability of exit increased dramatically, and those that

survived the imposition of sanctions lost revenues. However, their outcomes five years after the imposition of sanctions were not statistically different from those of firms that had previously exported very little food to Russia.

Figure 1. U.S. Food Exports to Russia and Control Countries, 2013–2019



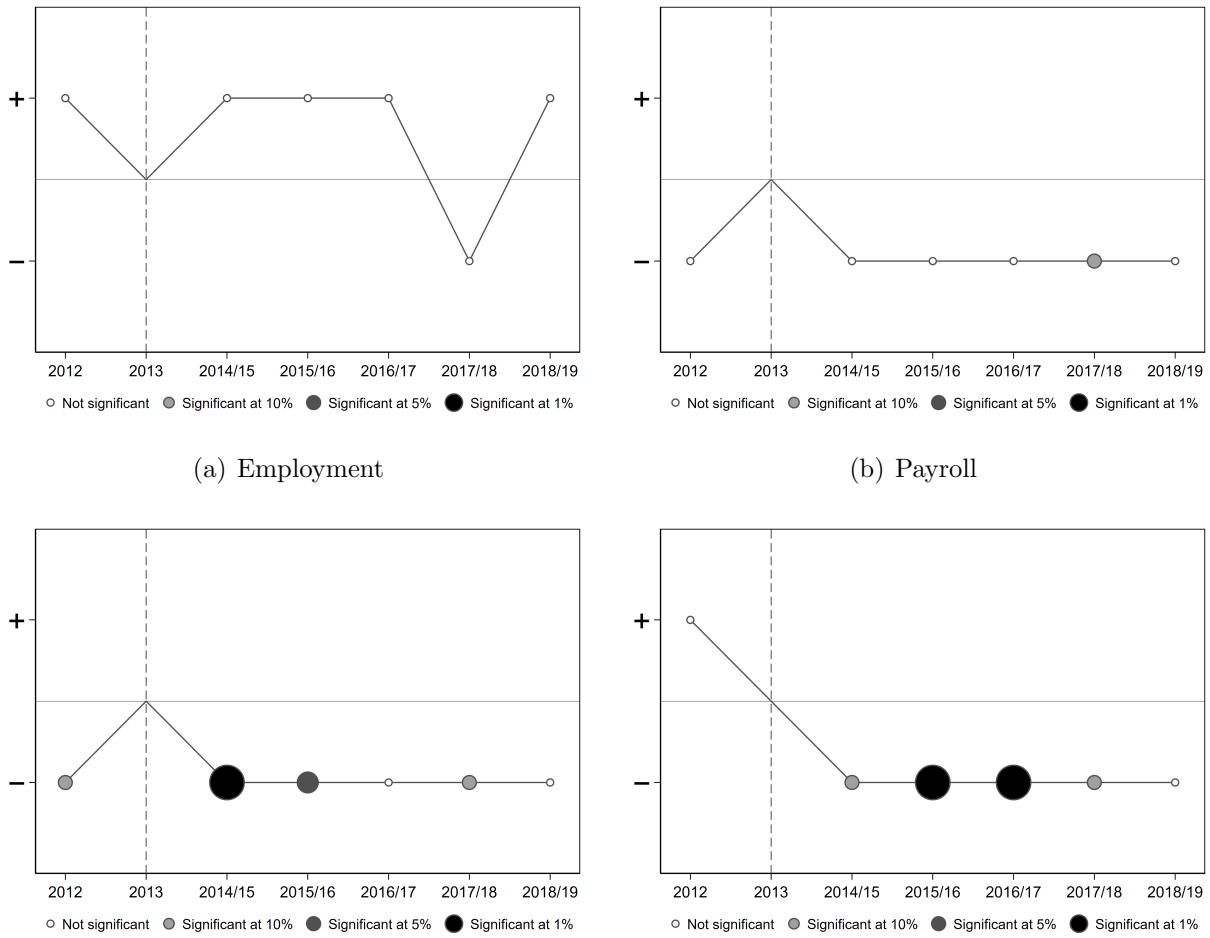
(a) Banned-Food Exports to Russia and Synthetic Control



(b) Other Food Exports to Russia and Synthetic Control

Notes: Real export values (in 2019 USD), not seasonally adjusted, of (a) food products subject to Russian import ban effective August 6, 2014 and (b) other food products not subject to the Russian import ban to Russia and a Synthetic Control, which is a weighted average of Angola (weight= 0.59) and China (weight= 0.41).
Source: Authors' calculations using public-use export flows (U.S. Census Bureau, 2022).

Figure 2. Impact of the 2014 Russian Food-Import Ban on U.S. Firm Activities
Event Study Dosage Regressions



(a) Employment

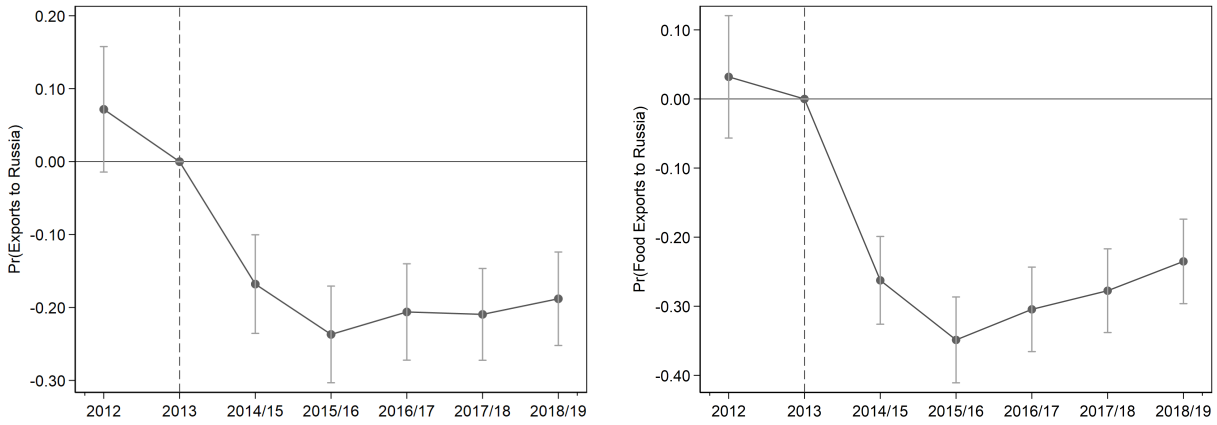
(b) Payroll

(c) Revenue

(d) Total Export Value

Notes: Sign and significance of regression coefficients from estimating Equation (4). All dosage regressions include year and firm fixed effects. Year 2013 is the reference year. Smaller and lighter markers indicate statistically insignificant coefficients; darker and larger markers indicate estimated coefficients that are statistically different from zero. Revenue regressions use weights to account for nonrandom missingness.
Source: Authors' calculations using LFTTD.

Figure 3. Impact of the 2014 Russian Food-Import Ban on U.S. Firm Exports
Event Study-in-Difference-in-Difference Regressions



(a) All Exports to Russia

(b) Banned-Food Exports to Russia

Notes: Regression coefficients from estimating Equation (2). Vertical bars represent 95% confidence intervals. All DDD regressions include destination by year and treatment status by year fixed-effect interactions. Year 2013 is the reference year.
Source: Authors' calculations using LFTTD.

Table 1. Summary Statistics of U.S. Firms Impacted by the 2014 Russian Food-Import Ban 2012–2013

Variable	Mean (SD)
Employment[a]	2,345
	(11,050)
Payroll[a,b]	128.3
	(580.0)
Revenue[a,b,c]	2,253
	(10,040)
Value of exports (all products, all destinations)[b,c]	131
	(713)
Prob(Exported to Russia)	0.78
Prob(Exported banned food to Russia)	0.75
Prob(Exported other products to Russia)	0.18
Prob(Exported banned food to other destinations)	0.86
Prob(Exported other products to other destinations)	0.61
Prob(Exported banned food to neighboring countries)[d]	0.22
Prob(Exported banned food to non-neighboring countries)[d]	0.86
Banned-food exports to Russia as share of total exports (Dosage)	0.15
	(0.28)

Notes: Summary statistics of treated firms in the pre-ban period. Means use annual data for 2012 and 2013. Standard deviations in parentheses for continuous variables.

[a] Conditional on nonzero value.

[b] Dollar amounts in millions of 2019 dollars.

[c] Weighted to account for nonrandom missingness.

[d] Neighboring countries include Belarus, Poland, Latvia, and Lithuania. Non-neighboring countries include all other countries, including Ukraine.

Source: Authors' calculations using LFTTD.

Table 2. Impact of the 2014 Russian Food-Import Ban on U.S. Firms: Export Behavior Regression Results

PANEL A: DDD Regressions

		Probability of Exporting...			
	Any Activity	Food to Russia	Other to Russia	Food to Elsewhere	Other to Elsewhere
β	0.0378*	−0.3018***	0.0674***	−0.0154	0.0187
	(0.0193)	(0.0191)	(0.0203)	(0.0189)	(0.0211)
Share outside $[0, 1]$[a]	0.0000	0.0044	0.0000	0.0000	0.0000

PANEL B: Dosage-Response Regressions

		Probability of Exporting Banned Food to...		
	Any Activity	Russia	Neighboring Countries[b]	Non-Neighboring Countries[b]
λ	−0.3821***	−0.0905*	0.0707**	−0.0450
	(0.0770)	(0.0527)	(0.0282)	(0.0657)
Share outside $[0, 1]$[a]	0.0000	0.0000	0.0000	0.0000

Notes: Regression coefficients from estimating Equations (1) and (3). All outcomes are binary. "Any Activity" is an indicator that equals 1 if the firm has any employment, revenue, or exports; these regressions include year fixed effects. All other regressions include both year and firm fixed effects. Robust standard errors in parentheses, clustered by firm. For brevity, column header "Food" refers to banned food, and "Other" refers to all other products.
* significant at 10%; ** significant at 5%; *** significant at 1%
[a] Share of predicted values that fall outside the unit interval.
[b] Neighboring countries include Belarus, Poland, Latvia, and Lithuania. Non-neighboring countries include all other countries, including Ukraine.
Source: Authors' calculations using LFTTD.

Table 3. Impact of the 2014 Russian Food-Import Ban on U.S. Firms: Firm Activity Regression Results

PANEL A: DDD Regressions				
	Employment	Payroll	Revenue[a]	Exports
β	0.0122	−0.0102	−0.0007	0.0994
	(0.0380)	(0.0431)	(0.0763)	(0.0823)

PANEL B: Dosage-Response Regressions				
	Employment	Payroll	Revenue[a]	Exports
λ	−0.0025	−0.2461	−0.6729*	−1.765***
	(0.1478)	(0.1905)	(0.3538)	(0.441)

Notes: Regression coefficients from estimating Equations (1) and (3). All outcomes are logged, and regressions also include year and firm fixed effects. Robust standard errors in parentheses, clustered by firm.
* significant at 10%; ** significant at 5%; *** significant at 1%
[a] Weighted to account for nonrandom missingness.
Source: Authors' calculations using LFTTD.

A Appendix

A.1 Sanctioned Products

Table A-1 lists the Schedule B codes targeted by the Russian sanctions. Combined, these products accounted for approximately 55% of the total value of U.S. exports of agricultural goods to Russia (Schedule B Chapters 1–24) in 2013 (authors' calculations from U.S. Census Bureau, 2022).

A.2 Selection of Synthetic-Control Countries

To select the synthetic-control countries, we use public-use exports at the ten-digit Schedule B level from the Census Bureau (U.S. Census Bureau, 2022). We use public-use data because the ban was imposed solely by Schedule B codes, and public-use statistics are available, since 2013, for U.S. exports by month and destination. We use monthly data for 2013 of log value of banned-food exports and the correlation between each destination country's vector of Schedule B shares of food exports and Russia's vector of Schedule B shares to select China and Angola as the synthetic-control countries.

Figure 1(a) in the main text shows the result visually. The dotted red vertical line marks July 2014, the last month pre-ban, but the synthetic control countries are selected using 2013 data only. The weights of 0.59 on Angola and 0.41 on China are not used in the analysis, except in Figure 1(a) and in Appendix A.3. Control set C2 includes all firms that exported banned food to China and/or Angola in 2012 and/or 2013 but did not export to Russia at all (so they are not in T or C1). Control set C3 includes firms that exported non-banned goods to China and/or Angola in 2012 and/or 2013, but are not in T, C1, or C2.

Table A-2 describes the four groups of firms used in the analysis, along with the count of firms in each group.

A.3 Aggregate Food Exports

To confirm that the decline in the value of exports of banned food to Russia shown in Figure 1(a) was driven by a decline in the quantity and not changes in prices, we examine the impact of the Russian import ban of agricultural products on aggregate U.S. exports of food products to Russia (treated destination) at the product-country-year level, where a product is a ten-digit Schedule B code, using public-use export flows (U.S. Census Bureau, 2022). We use the same control countries as in the main text, China and Angola; we use the weights derived in Appendix A.2 to create a single synthetic-control country for this analysis. This approach is different from the one we take in the paper, where we do not use the synthetic-control weights.

We estimate the following difference-in-differences (DD) specification,

$$\mathrm{Y}_{pct} = \alpha_{pc} + \tau_t + \gamma \cdot \mathrm{Post}_t \cdot \mathrm{Russia}_c + \varepsilon_{pct} \tag{A-1}$$

where p denotes a ten-digit Schedule B product, c denotes a destination (Russia or synthetic-control country), and t denotes year. The product-country fixed effects α_{pc} absorb product-by-country, time-invariant differences in the outcome variable Y_{pct}; τ_t controls for common, time-varying, macroeconomic impacts; and the Russia_c indicator absorbs country-specific differences. The coefficient γ captures the differential change in the outcome variable for banned-food exports to Russia in the post-ban period.

The outcome variables Y_{pct} we examine are log value, log quantity, and log unit value. Unit value is calculated as the ratio of value to quantity. For homogeneous products, including agricultural products, this ratio closely approximates price (Fast, Fleck, and Smith, 2022).

The γ coefficients from the DD regressions are reported in Table A-3 for the three outcomes of interest. Panel A shows the γ coefficients estimated using banned-food products only. For comparison, we repeat the analysis using exclusively non-banned-food products in

Panel B. In both panels, we find that the value and quantity exported declines. However, the treatment effect for exports of banned-food products is three times larger than the treatment effect for non-banned-food products. In both cases, changes in unit values are not statistically significantly different from zero.

These patterns confirm that the Russian food-import ban halted nearly all sales of the banned-food products to Russia. However, the aggregate data do not allow an examination of individual exporters' responses to the ban with respect to their ability to make up for the lost sales to non-Russian destinations, diversifying their product mix away from the banned products, or altogether going out of business. We study these firm-level adjustments using confidential microdata.

Table A-1. Banned Foods Products, Schedule B Codes

Schedule B Code	Description
0201	Meat of cattle, fresh and chilled
0202	Meat of cattle, frozen
0203	Pork fresh, chilled or frozen
0207	Meat and food byproducts of poultry
0210	Meat salted, pickled, dried or smoked
03	Fish and crustaceans, mollusks and other aquatic invertebrates
0401–0406	Milk and dairy products
07	Vegetables, edible roots, and tuber crops
0801–0811, 0813	Fruits and nuts
160100	Sausages and similar products from meat, meat byproducts or blood
190190	Prepared products including cheeses and curd on the basis of vegetable oils
210690	Food preparations not elsewhere classified, containing milk, based on vegetable oils

Notes: Food products by Schedule B codes included in Russia's import ban.
Source: U.S. Department of Agriculture (2014a).

Table A-2. Treatment and Control Groups, DDD Analysis

	2012–13 Exports to...	
	Russia	China/Angola
Banned Food	**T** (Firms=300)	**C2** (Firms=1,600)
Other products	**C1** (Firms=6,300)	**C3** (Firms=40,000)

Notes: All counts are rounded to comply with Census Bureau rules on disclosure avoidance. Groups C1, C2, and C3 exclude firms in T. Groups C2 and C3 exclude firms in C1. Group C3 also excludes firms in C2.
Source: Authors' calculations using LFTTD.

Table A-3. Impact of Russian Food-Import Ban on U.S. Food Exports to Russia
Difference-in-Differences Results Using Public Data

	PANEL A: BANNED FOOD		
	Value	Quantity	Unit Value
γ	-1.1579^{***}	-1.2235^{***}	0.0656
	(0.3120)	(0.3140)	(0.0851)
Observations	2,699	2,699	2,699

	PANEL B: OTHER FOOD		
	Value	Quantity	Unit Value
γ	-0.3376^{**}	-0.3816^{***}	0.0440
	(0.1323)	(0.1445)	(0.0586)
Observations	5,218	5,218	5,218

Notes: Regression coefficients from estimating Equation (A-1) at the product-country-year level on the sample of food products subject to the ban (Panel A) and food products not subject to the ban (Panel B). All outcomes are logged. All regressions include year and destination by ten-digit Schedule B fixed effects. The synthetic control group is a weighted average of Angola (weight= 0.59) and China (weight= 0.41). Robust standard errors in parentheses, clustered by ten-digit Schedule B code.
* significant at 10%; ** significant at 5%; *** significant at 1%
Source: Authors' calculations using public-use export flows (U.S. Census Bureau, 2022).

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